

Fall-Detection Engineering Decision Matrix

Model	Accuracy 35%	Recall 25%	Runtime Efficiency 15%	Mobile Deploy. 15%	Interp. 10%	Weighted Score
Gradient Boosting	4.96	5.00	4.00	4.00	4.00	4.586
SVM (RBF)	4.89	4.92	4.50	4.00	3.00	4.517
Random Forest	4.95	4.99	3.00	3.00	4.00	4.281
Extra Trees	4.96	4.99	3.00	3.00	3.50	4.233

Scores are normalized on a 1-5 scale; final score is the weighted sum.